Measurements of Impacts of Scientific Infrastructure Constructions on Innovators

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Introduction

We are trying to understand the effects of having exposure to the construction of visible important scientific construction on the existence of inventors. The rationale is that once people observe these constructions they are more willing to pursue careers in science and technology, becoming what we will call "Inventors".

Databases

The databases can be divided in two types, the database that comes the inventor data and the other databases that provide information on population, for example, such that we can build relative measures and not absolute ones.

The Inventors Database

The first main database is the SciencePo Wikipedia Webscrapping database. You can find it at: https://data.sciencespo.fr/dataset.xhtml?persistentId=doi:10.21410/7E4/RDAG3O. More details are in this paper: https://www.nature.com/articles/s41597-022-01369-4. In the code we download directly, so there is no need to download this. This database, basically, scraps all people with detailed data on Wikipedia. We select only people from 1800 on and in the Science/Discovery category, which is our focus now.

The Population Databases

For the country analysis, we also used the Gap Minder database, which can be found here: https://www.gapminder.org/data/. This data also downloads directly in the R code. We use the cross country data for "New Births". In this we have yearly data.

The first attempt of using more granular data is using HYDE. This database basically divides the world into 85 km2 cells and attributes a population for each cell for each decade. We use it only from 1800 onwards. This database has to be downloaded and it is very large, it can be found here: https://landuse.sites.uu.nl/datasets/. We used the HYDE 3.1 due to the size of the files and what we actually need.

Finally, the most recent approach focuses only in the US and it uses county level population data. This data comes from: https://data2.nhgis.org/main and I downloaded by hand selecting the Total Population at the County Level for all years in the time series type of data. Basically, this data gives the population by county and decade since 1850 (because it uses Census data).

Facilities

We constructed the Facilities through research and ChatGPT use. Basically, it provides information on major constructions in the end of 19th century and beginning of the 20th century. We use the coordinates to give the exact location of each infrastructure construction and from the Wikipedia database we create a variable of construction start year to assign the treatment group. Most Wikipedia pages give an exact idea of when this constructions started, however there are some examples that are not very clear, excluding these cases originating from the "selected" page in the spreadsheet. You can find the constructions here: scientific_facilities.xlsx.

Chronology and Rationale

First, we began by using country level population data. To do this analysis we used the latitude and longitude of birth of Inventors (this is present in the Wikipedia Database) and birth year as well. Thus, we could assign each inventor to a year and a country. We aggregated at the decade level (and also used an alternative decade specification). We then ran event studies with this data. The event studies used that a country becomes treated after the construction of a major scientific building and it remains treated afterwards. The outcome

variable was the relative number of inventors, precisely the rate of inventors on 100k population.

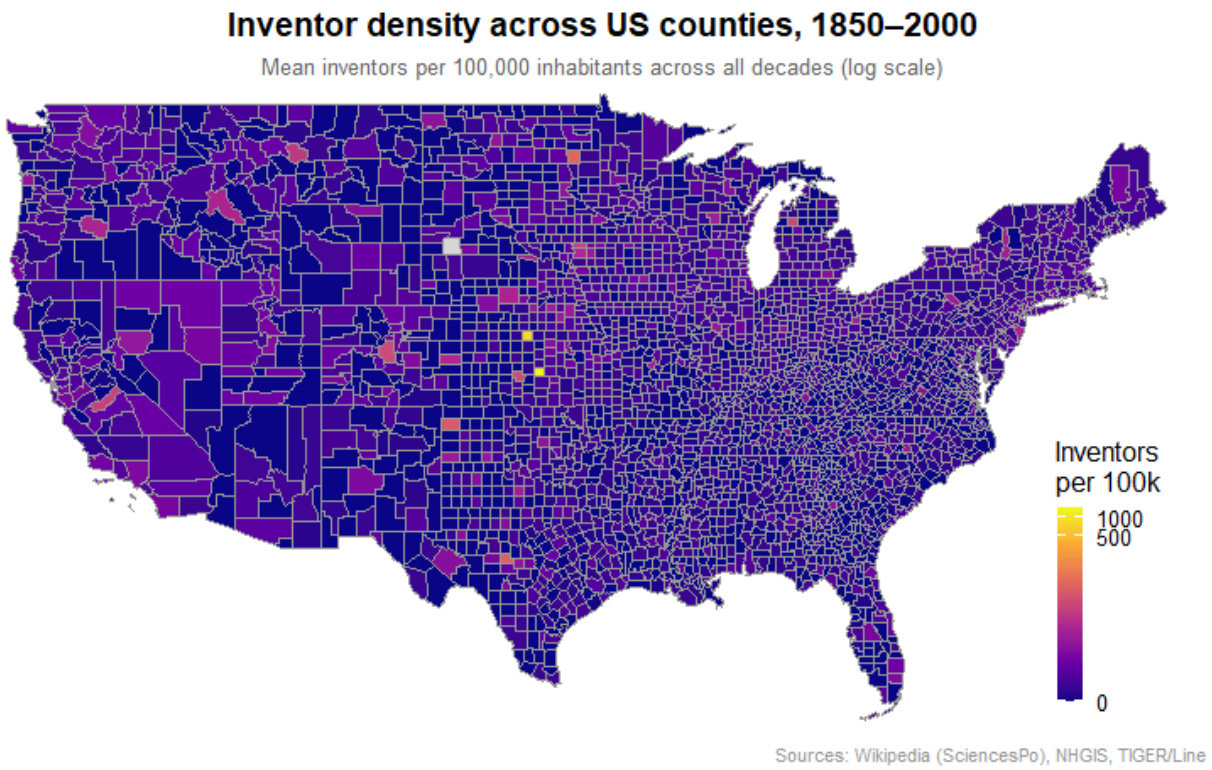
This analysis can have many flaws, the first is that the construction is not an exogenous event, probably comes bundled with more investment in other formats. However, for our goal, the most important issue is that the whole country becomes treated with an intervention that is very local, not everyone in a country will be able to see this construction and thus the mechanism we expect is not being captured. Thus, we moved to a more granular analysis using the HYDE database.

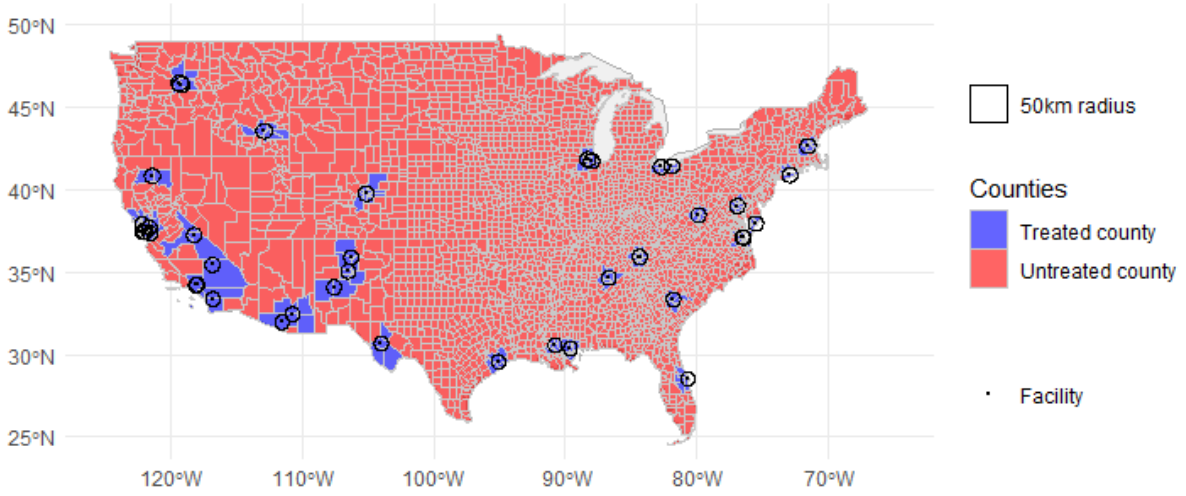
Since HYDE gives a very precise (granular) population data we can do a very different analysis. We use the coordinate location of birth of each inventor to assign each inventor to a cell in the world. From this we can construct our measure of inventors per 100k with this new granular population data. Thus, we create a panel of cell X decade for this outcome variable. We use the coordinates of the construction to determine treatment, with the rationale that cells near the construction should be treated. Thus, a cell becomes treated based on the year of construction and if the center of the cell is 50km distant from the construction. The remaining cells are considered controls.

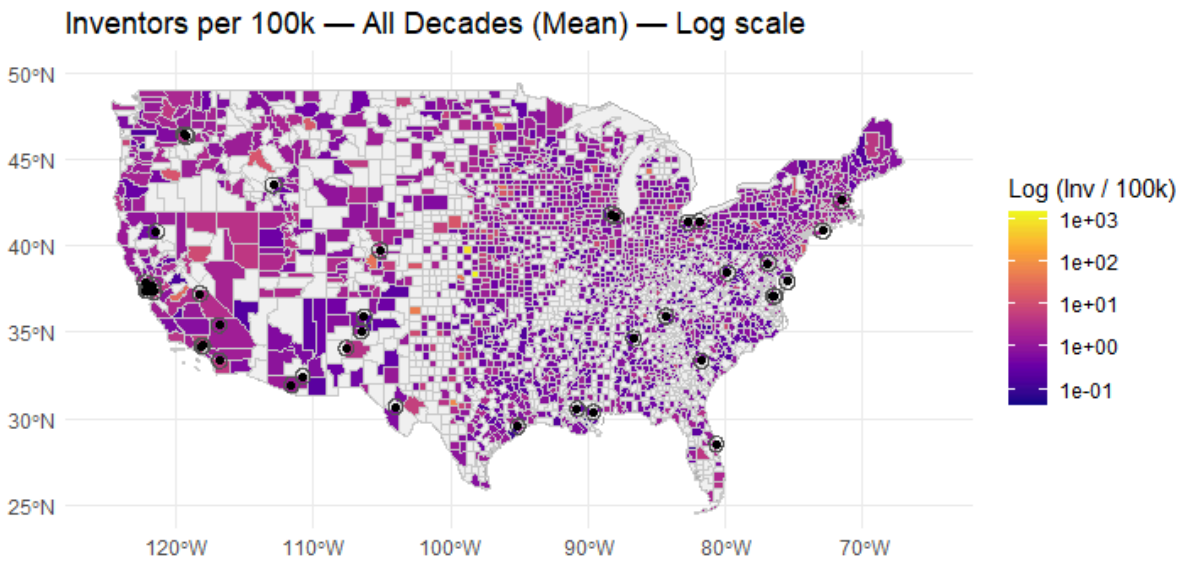
Even though this made more sense, we decided to focus on the US to control some kind of institutional exposure, which is quite reasonable. We chose the US because it has enough facilities, geographical variation and a lot of inventors. The analysis using HYDE for the US is very similar to before but using just the US data, for treated and control groups.

Finally, we reach this last exercise in which we do something very similar but for the county data and not HYDE. Now we don't have a 85 km2 cells, but we have the county limits, such that we can assign each inventor to each county. Thus we can construct the county x decade panel of inventors per 100k population. The only difference is that now, since counties are larger, we don't use the center of the county to determine treatment, we use it if it reaches the limit of the county.

County Level Analysis

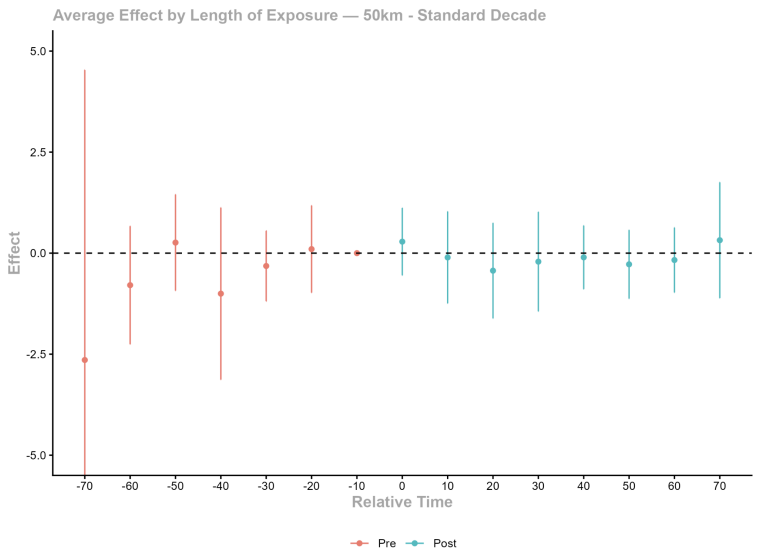

Inventor density across US counties, 1850–2000
Mean inventors per 100,000 inhabitants across all decades (log scale)
Inventors per 100k
1000
500
0
Sources: Wikipedia (SciencesPo), NHGIS, TIGER/Line

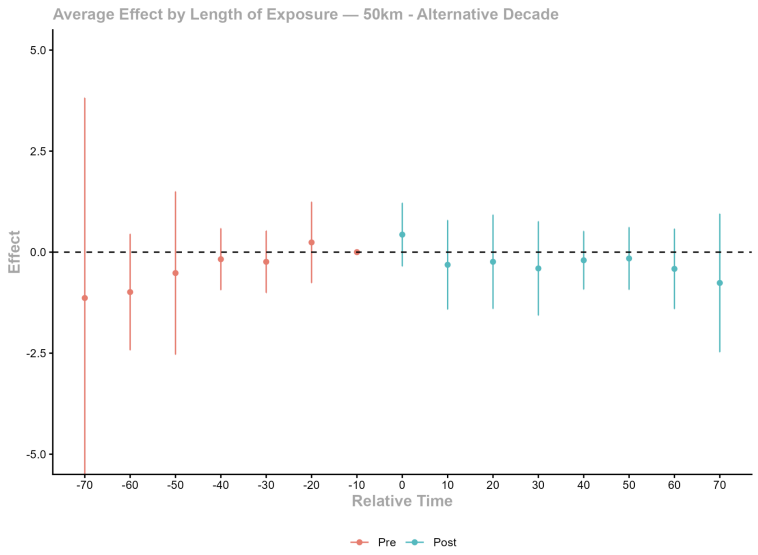

50°N
45°N
40°N
35°N
30°N
25°N
120°W
110°W
100°W
90°W
80°W
70°W
50km radius
Counties
Treated county
Untreated county
Facility



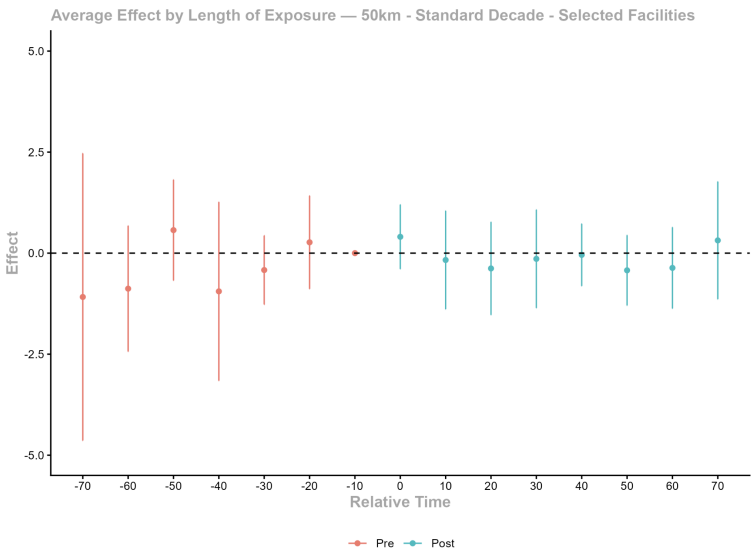
Event Studies

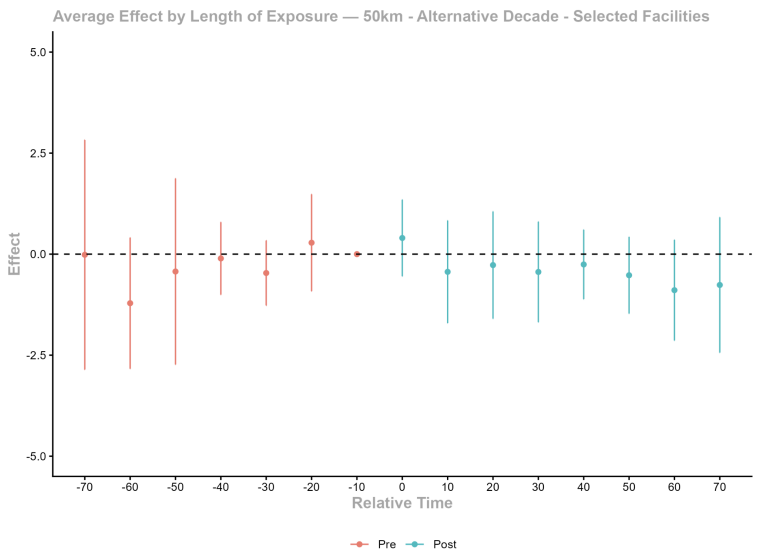
Full Database





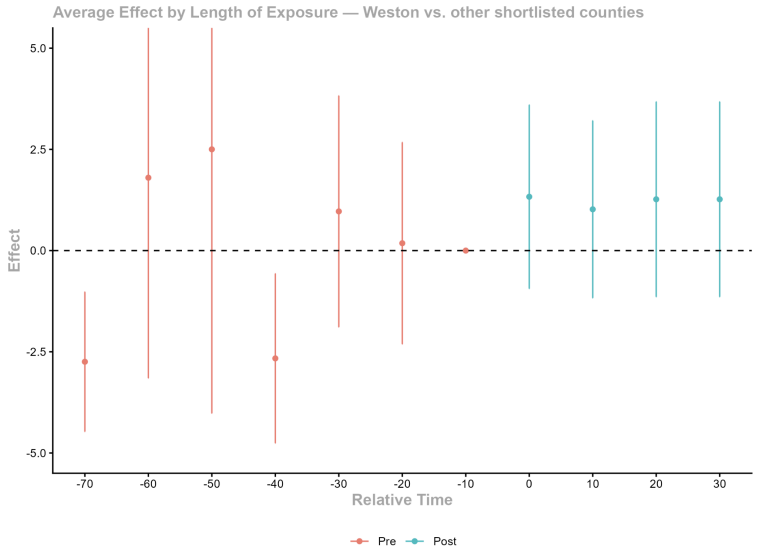
Selected Database



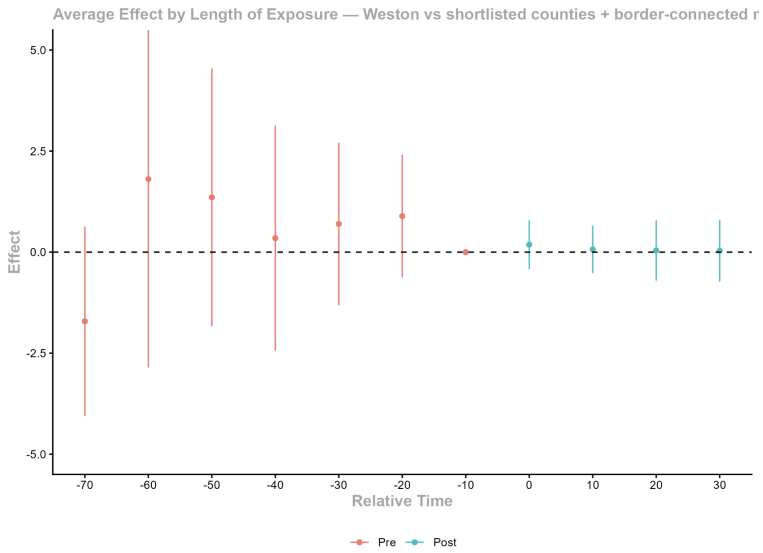


Event Study with Fermilab

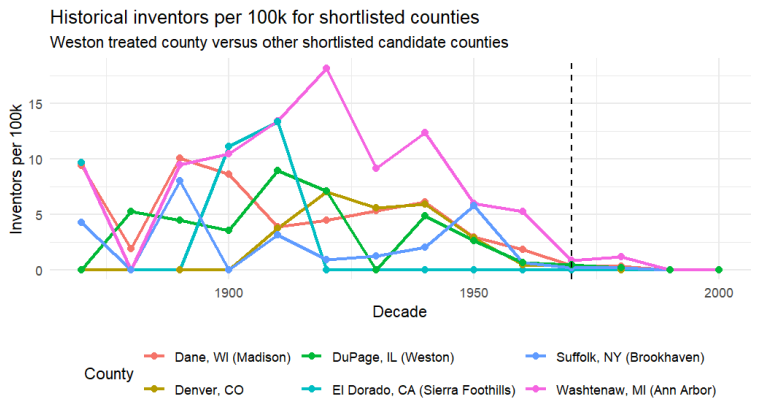
In this exercise we use that many places were considered to be the host of what later would be known as the Fermilab, here we use the Weston County as treated and the other counties in the final list as control. The result follows:



In another specification, I considered as treated the Weston county and all the counties that have a border with it. As control, I considered the other 5 considered places and also their "neighbor"(share a border) counties:



We can also look at the historical series for this 6 places and it appears that one of the reason we can attribute this little effect is because almost all of the counties converge to 0, due to the timing, many people born in 1970 or 1980 may not be a known inventor.



The Top Counties

The top counties of rate of inventors are:

1. Rice County, Kansas. Became a large hub for agricultural innovation, and rail line that passed through
2. Smith County, Kansas. Another county associated with agricultural development, and another possibility being the exploration of oil in the Kansas region.
3. Clay County, Minessota. Associated as well with agricultural innovation and railway expansion.
4. Menard County, Texas. No clear explanation to why this would have a high rate of scientists.
5. Dallam County, Texas. No clear explanation. One possibility is that counties in Texas had inventors in some earlier industries like oil or energy.